Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

CSE4204 — Mobile Computing Lab · Final Project Presentation

RESUMIND

AI Resume Analyzer

Presented by — Team CSE4204-8A-T05 (Section 8A)

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Supervised by

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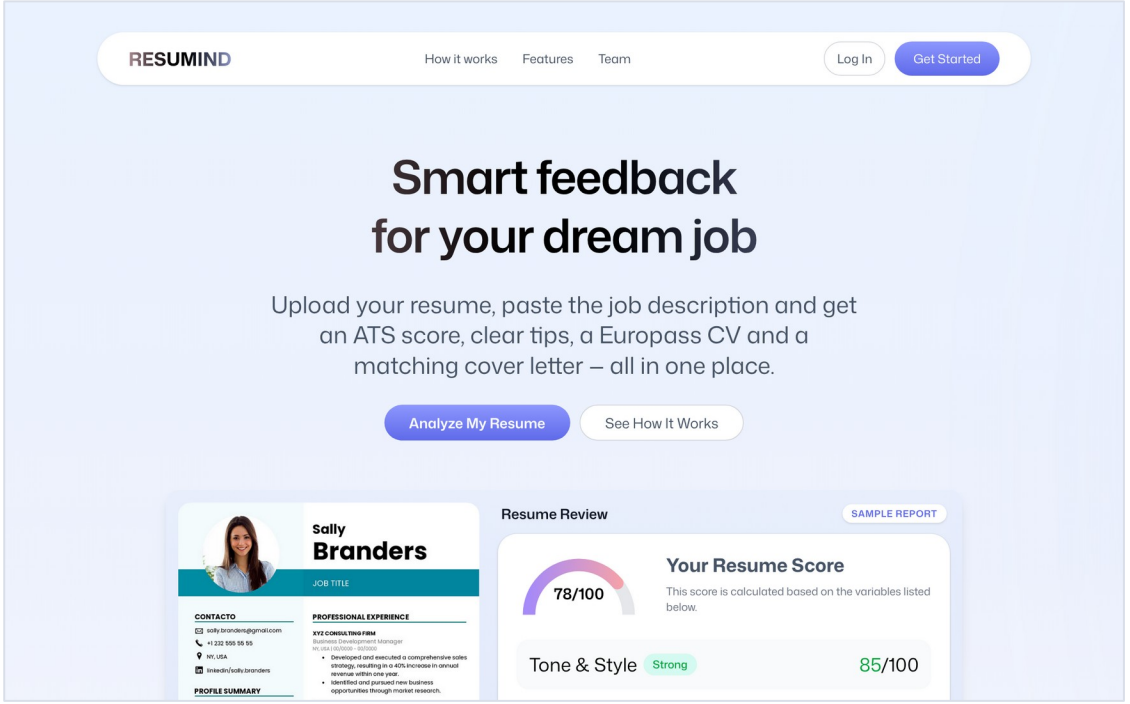
Assistant Professor, Dept. of CSE

NUBT Khulna

Project Introduction

RESUMIND is a web application that checks a resume for a job and helps the user improve it.

- Upload a resume (PDF) with the job details
- Get an AI score and simple tips
- Get an improved resume in Europass format
- See the score before and after improving
- Create a cover letter for the same job
- Works on phone, tablet and computer



Problem Statement

Filtered by ATS

Many resumes are rejected by software before a person reads them.

No feedback

Applicants do not know what is wrong in their resume.

Unknown standard

Many jobs abroad want a Europass CV, which most students do not know.

Time-consuming

Writing a new cover letter for every job takes a lot of time.

Existing tools: most online resume checkers are paid or do only one part of the work.

Objectives

O1 Safe sign up and login

O2 Save every resume and result

O3 Check a resume with AI and give a score

O4 Improve the resume in Europass format

O5 Show the score before and after

O6 Create a cover letter for the job

O7 Easy to use on phone and computer

O8 Test the system and put it online

Key Features

1. Upload

Resume PDF and job details

2. Analyse

AI score with tips

3. Europass CV

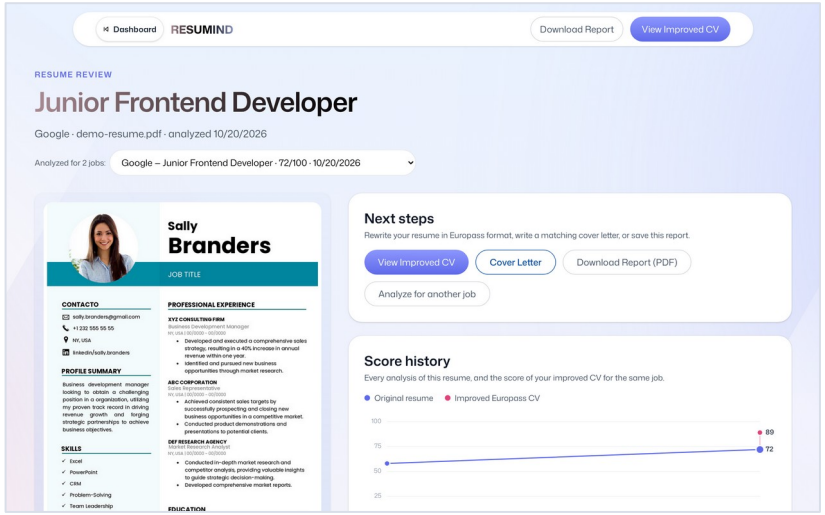
Improved CV you can edit

4. Before / after

See how much it improved

5. Cover letter

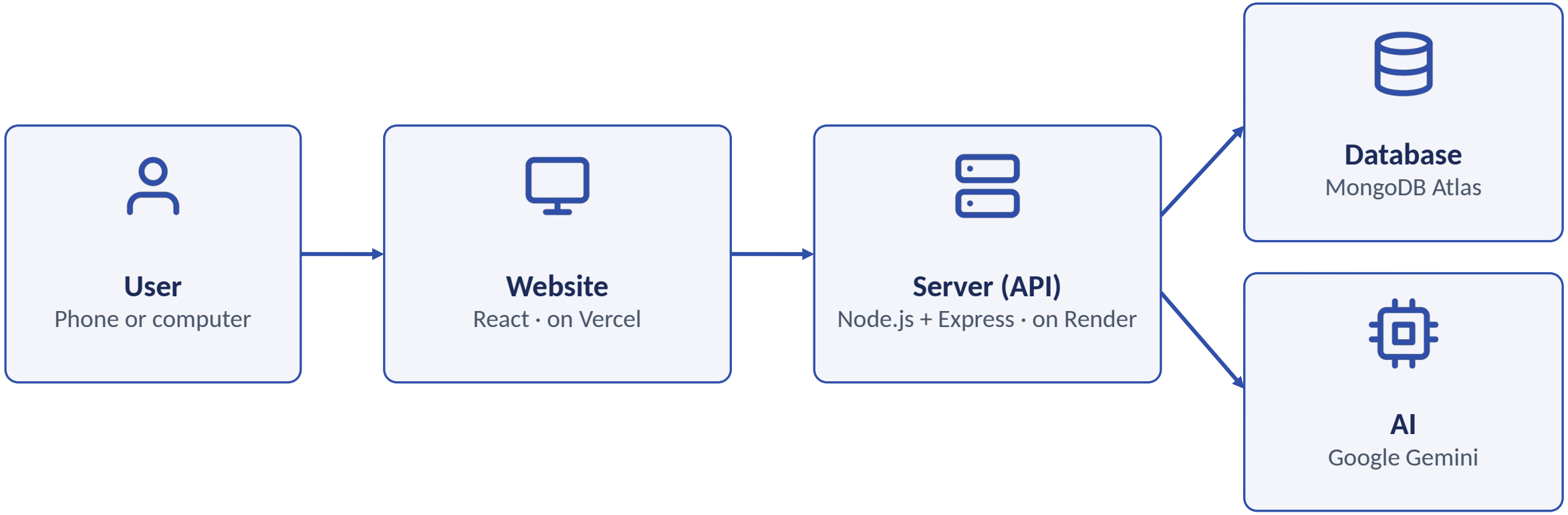
One-page letter for the job



Also included

- Dashboard with all past results
- Login with email or Google
- Download as PDF
- Works on phone and computer

System Architecture



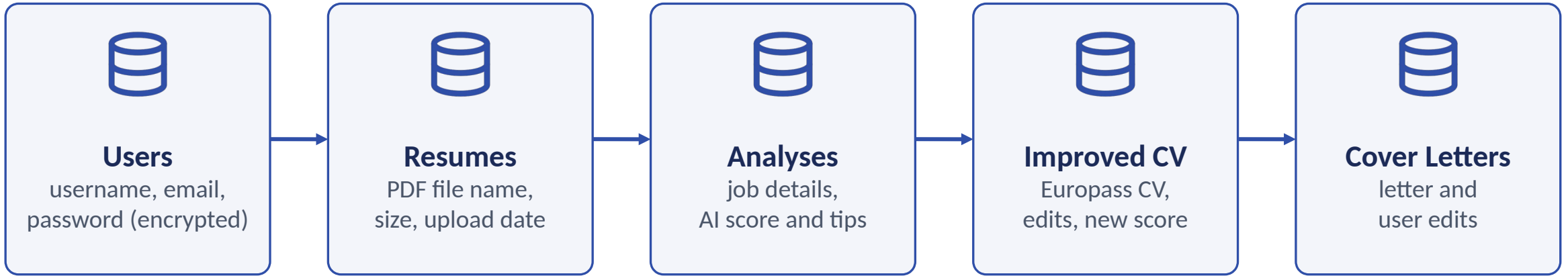
How it works: the user opens the website, the website sends the request to our server, and the server saves data in MongoDB and asks Google Gemini for AI results. Secret keys stay only on the server.

Technology Stack

Part	Technology Used
Frontend	React, TypeScript, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB Atlas
AI	Google Gemini API
Security	bcrypt (password), JWT (login), Google Sign-In
Testing	Vitest, GitHub Actions
Hosting	Vercel (frontend), Render (backend)
Tools	VS Code, Git and GitHub

Database Design

We use MongoDB Atlas (cloud database). Data is stored in 5 collections:



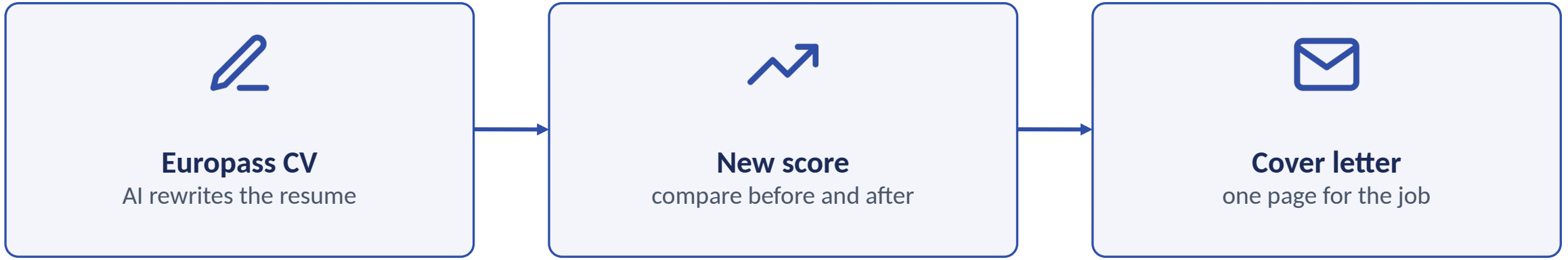
- One user can have many resumes; one resume can be analysed for many jobs.
- Each analysis can have one improved CV, and each improved CV one cover letter.
- The PDF files themselves are stored in MongoDB GridFS.
- Every record stores its owner, so each user sees only their own data.

AI Workflow

Step 1 — Analyse the resume



Step 2 — Improve (one click each)



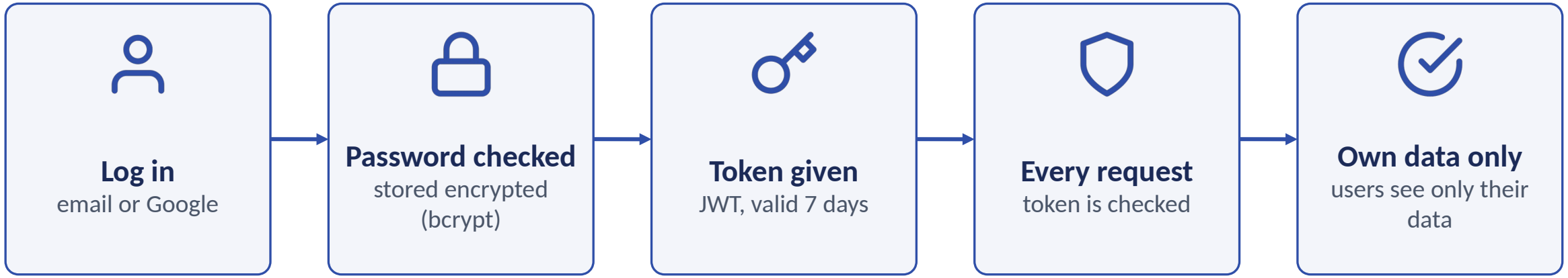
Keeping the AI Correct

AI can sometimes add wrong information. Our code checks the improved CV and fixes it:

AI wrote	After our check
A number that is not in the resume ("40%")	[Add a measurable result] — the user fills it in
November 30, 2024	11/2024 (Europass date format)
H.S.C, 4.67	Higher Secondary Certificate, GPA 4.67 / 5.00
Religion, date of birth	Removed (not used in Europass)

So the AI never adds false information to the CV.

Login and Security



Other security

- Too many wrong logins are blocked for a while
- Changing the password logs out other devices
- Only PDF files are accepted (max 10 MB)
- Secret keys are never in the code

API part	What it does
Auth	Sign up, login, change password
Resume	Upload, view and delete resumes
AI	Analyse, improve CV, cover letter

Testing and Deployment

48

automatic tests, all passed

- Login, upload and AI features are tested
- Tests run every time code is pushed to GitHub
- Also tested by hand on phone and computer

Vercel — Frontend

ai-resume-analyzer-psi-lime.vercel.app

Render — Backend API

resumind-api-3kpd.onrender.com

MongoDB Atlas — Database

stores users, resumes and results

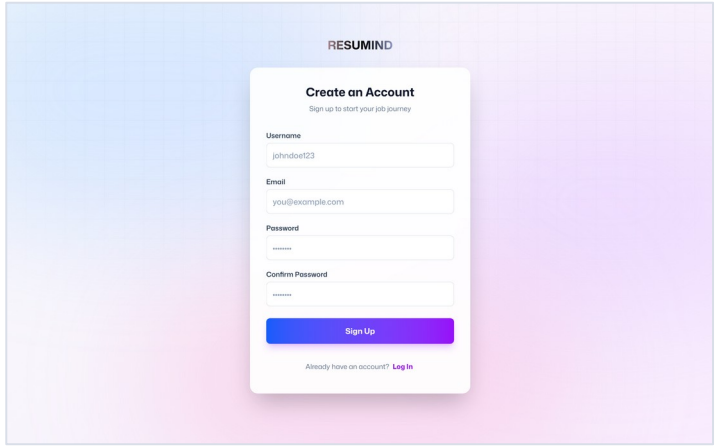
GitHub — source code

updates go live automatically

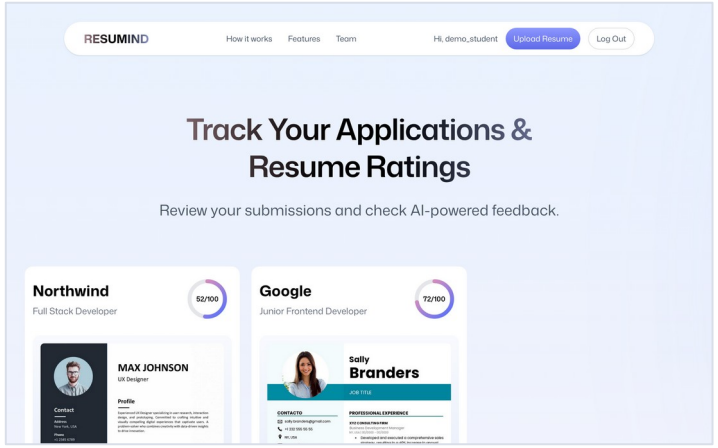
Scan to open the live app

Secret keys are kept in .env files, not in the code.

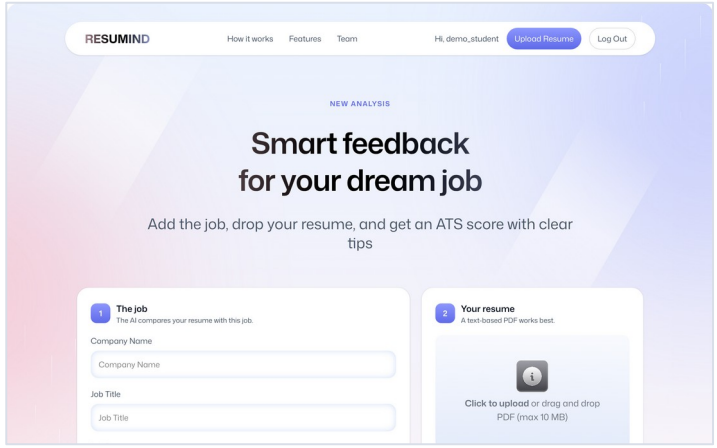
Project Screenshots



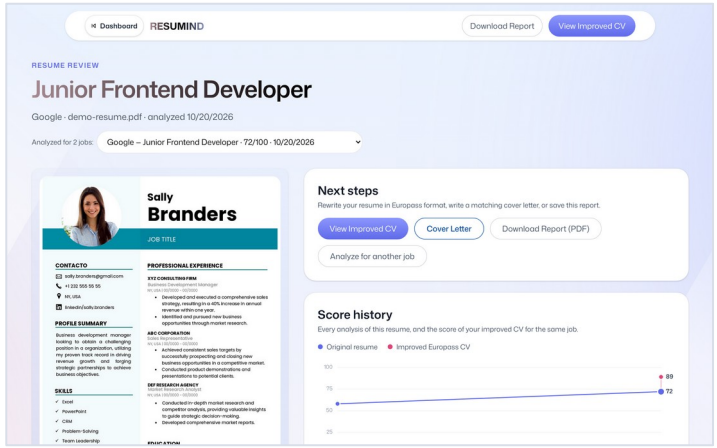
Login / Sign up



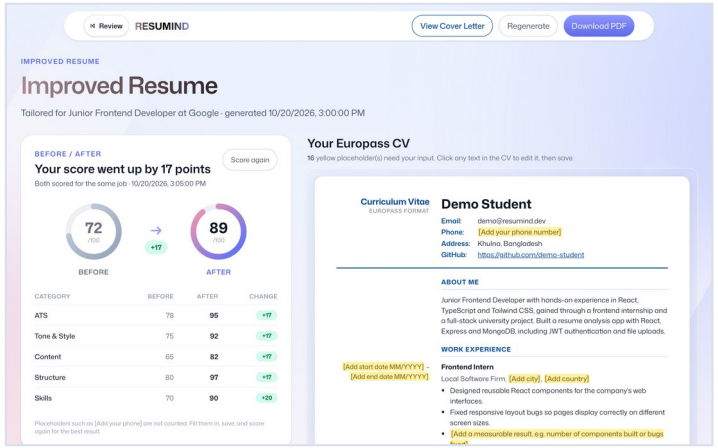
Dashboard



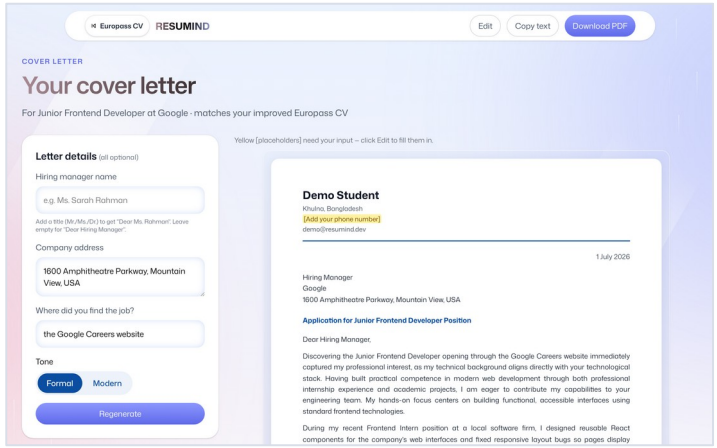
Upload resume



Analysis result



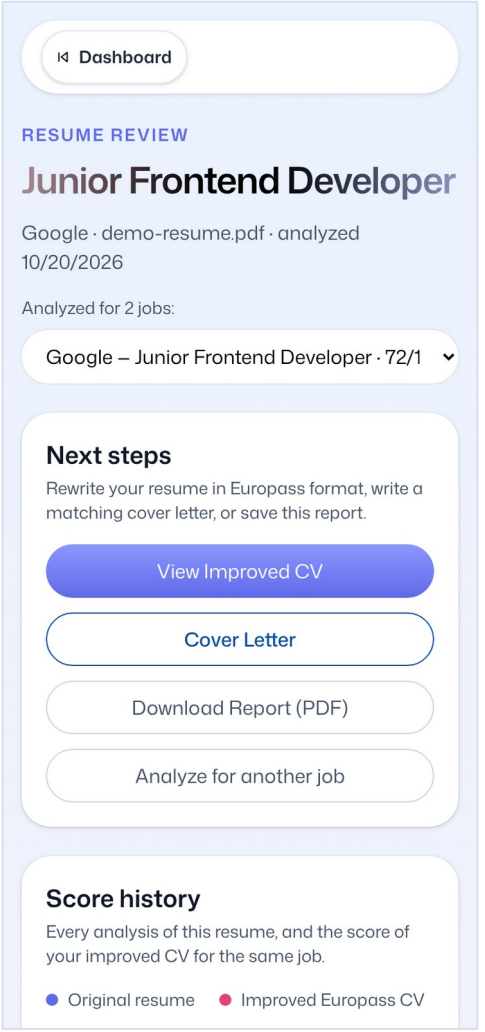
Europass CV



Cover letter

Live Demonstration

1. Open the live site on laptop and phone
   Mahammad
2. Sign up / Sign in with Google
   Ratul
3. Upload a resume for a real job
   Sakib
4. Read the scores and tips
   Sakib
5. Create the Europass CV
   Sakib
6. Score the improved CV
   Mahammad
7. Generate and download the cover letter
   Mahammad
8. Show the dashboard and deployment
   Himel & Ratul



Challenges and Solutions

Challenge	How we solved it
AI answers came in different formats	Asked the AI for a fixed format and checked it in code
AI sometimes added wrong information	Our code removes it and asks the user to fill in
AI server was sometimes busy	The app tries again automatically
Uploaded files were lost after updates	Files are now saved in the database
Big pages were hard to use on phones	Designed for mobile first

Future Improvements

Scanned resumes

Read photo/scanned PDFs and Word files

Job link

Paste a LinkedIn or BDJobs job link

Interview practice

AI asks practice questions

Bangla support

Interface and feedback in Bangla

Mobile app

Android and iPhone app

For recruiters

Compare many resumes for one job

Thank You

Questions?



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